

LLM: Training Model

Training: modelni parameterlarini uzgartirishi orqali uni bashoratlarini yaxshilash jarayon

Model training bilan **Model building** bitta jarayon emas. Birinchi modelni fundamentini yasavolamiz (data preprocessing, tokenizer, embeddings, etc.). Keyin, ushani training qilamiz.

Self Supervised learning: input qiymatlar biz tomondan beriladi lekin **output qiymatni uzi** dataset builderni ichida **yasaydi**.

```
class Trainer:
    def __init__(self, model, dataset_builder, train_bin, val_bin,
                  model_config, train_config, device):
        self.model = model.to(device)
        self.dataset_builder = dataset_builder # self supervised learning
```

Loss: modelni next word prediction i haqiqiy word bilan **qanchalik yaqin bulganini** chiqarib beradi. loss qiymati oz bulgani sari yaxshi.

Working principle: high loss -> backpropagation -> found new weights (W, b) -> Optimizer (AdamW) updates weights -> loss decrease

```
# ---- optimizer ----
weight_decay: float = 0.1
beta1: float = 0.9
beta2: float = 0.95
grad_clip: float = 1.0
```

LR (Learning Rate): urganish qadami bulib juda kichkina yoki katta berilmaydi. Birdaniga kutarmasdan warmup (qizitib) qilib keyin cosine decay bilan birga ishlatish.

```
# ---- schedule ----
max_steps: int = 3000
warmup_steps: int = 100

lr_max: float = 3e-4
min_lr_ratio: float = 0.1
```